

### 1.2.2 Use of apparatus and techniques

| Learning outcomes                                                                                                                                                                                                                                                               | Additional guidance |
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| <i>Through use of the apparatus and techniques listed below, and a minimum of 12 assessed practicals (see Section 5g), learners should be able to demonstrate all of the practical skills listed within 1.2.1 and CPAC (Section 5g, <b>Table 2</b>) as exemplified through:</i> |                     |
| (a) use of appropriate analogue apparatus to record a range of measurements (to include length/distance, temperature, pressure, force, angles and volume) and to interpolate between scale markings                                                                             | HSW4                |
| (b) use of appropriate digital instruments, including electrical multimeters, to obtain a range of measurements (to include time, current, voltage, resistance and mass)                                                                                                        | HSW4                |
| (c) use of methods to increase accuracy of measurements, such as timing over multiple oscillations, or use of fiduciary marker, set square or plumb line                                                                                                                        | HSW4                |
| (d) use of a stopwatch or light gates for timing                                                                                                                                                                                                                                | HSW4                |
| (e) use of calipers and micrometers for small distances, using digital or vernier scales                                                                                                                                                                                        | HSW4                |
| (f) correctly constructing circuits from circuit diagrams using DC power supplies, cells, and a range of circuit components, including those where polarity is important                                                                                                        | HSW4                |
| (g) designing, constructing and checking circuits using DC power supplies, cells, and a range of circuit components                                                                                                                                                             | HSW4                |
| (h) use of a signal generator and oscilloscope, including volts/division and time-base                                                                                                                                                                                          | HSW4                |

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| <b>(i)</b> | generating and measuring waves, using microphone and loudspeaker, or ripple tank, or vibration transducer, or microwave/ radio wave source | HSW4       |
| <b>(j)</b> | use of a laser or light source to investigate characteristics of light, including interference and diffraction                             | HSW4       |
| <b>(k)</b> | use of ICT such as computer modelling, or data logger with a variety of sensors to collect data, or use of software to process data        | HSW3, HSW4 |
| <b>(l)</b> | use of ionising radiation, including detectors.                                                                                            | HSW4       |

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